Using Digital Twin Simulation for Synchronous Communication Research: With an Application to the Structured Group Design*

Kevin M. Esterling
University of California Riverside

Ben Treves
University of California Riverside

Kelton Adey
University of California Riverside

Euchan Jang
The Ohio State University

August 10, 2026

*Previous versions were presented at at the 2026 Annual Meeting of the Western Political Science Association, the 2025 Annual Meeting of the American Political Science Association, the UCR Mass Behavior Workshop, and the February 2026 LOGOS Open Day Conference at UCR. We thank Lynn Vavreck for suggesting our exploration of focus groups, and Jessica Feezell, Jieun Lee, and Michelle Torres for extremely helpful comments on the paper. The Python scripts needed to run the simulations described in this paper are available at the project GitHub repository. The data and code needed to reproduce the reported results are available at the reproduction GitHub repository. See the appendix for our AI Disclosure. Correspondence to: Kevin M Esterling, University of California Riverside, 900 University Ave., Riverside, CA 92521 USA, kevin.esterling@ucr.edu.

Abstract

We use LLM digital twin simulation to model human group interaction under alternative synchronous research design choices, at scale and across repeated trials – revealing the sampling distribution of the qualitative results under each design. These methods generalize to any synchronous communication design, and enable us to state the assumptions required for the simulation to add to the accumulation of knowledge as well as validation methods that can help weaken these assumptions. The software is open source, and easy to use and modify. We use these simulation methods to evaluate a novel large-N *structured group design* under which text data remains stationary across repeated trials and inferentially captures diverse perspectives and reasoning within populations.

KEYWORDS: Digital Twins Simulation, Large Language Models, Algorithmic Curation, Synchronous Design, Deliberation, Focus Groups

1 Introduction

Human opinions are formed through communication in social interactions (Chambers, 2018; Habermas, 1984). Learning how opinion emerges from social interactions requires synchronous research designs that collect closed- and open-ended data while participants actively engage each other in discussion (Esterling et al., 2021). Engaging in group-based interaction can lead participants to access deeper cognitive processes (Esterling et al., 2011; Tetlock, 1983) rather than just offering top of the head responses like in a standard survey (Zaller and Feldman, 1992).[1] The design or context for those interactions, however, heavily conditions the nature and dynamics of discussion (Cyr, 2019; Dryzek et al., 2019; Fishkin, 2009; Niemeyer et al., 2024), and so there is a need to understand the *properties* of alternative discussion designs for evaluating emergent group interaction.

An ideal method for evaluating the properties of synchronous designs would require sampling participants and using a given experimental design many times over, across different designs, each time with human participants. This procedure would reveal the sampling distribution of results generated under each design. Using human participants in synchronous designs, however, is extremely expensive and time consuming – particularly when the design scales to require hundreds or thousands of participants that must meet at the same place and time.[2] Here we develop low-cost methods for digital twins LLM simulation to explore the emergent properties of group discussion under different discursive contexts and designs, and at arbitrary scale (Wang et al., 2025; Zhang et al., 2024). We specify validation tests and state the assumptions required for digital twins simulation methods to contribute to the accumulation of knowledge for discursive design.

[1] Asynchronous platforms cannot replicate the dynamics of synchronous human discursive interaction as they tend to reduce spontaneity, shorten answers, and create uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

[2] To give a sense of the costs, the "Connecting to Congress" Deliberative Town Hall experiments (Neblo et al., 2018) used probability sampling to recruit participants for online synchronous meetings, and each session of 20-30 participants cost about $10K to host. The AmericaSpeaks "Our Budget, Our Economy" synchronous in-person deliberation event (Esterling et al., 2021) recruited over 2,000 participants, and the project required a year of planning and a budget of over $2M.

The digital twins methods we introduce are general to any synchronous design. Here we introduce and evaluate the *structured group design*, which is a novel design for human synchronous discussion groups that can scale to any size group. The structured group design enables synchronous human participant input at scale through a three-stage discussion format centered on a substantive topic, motivated by a closed-ended survey question, relying on an LLM pipeline to capture diverse perspectives submitted simultaneously by all participants in real time during a synchronous event (similar to Lu et al., 2024). The structured group design contrasts with traditional synchronous designs where only one participant can speak at a time, such as deliberation experiments (Esterling et al., 2021; Neblo et al., 2018) or focus groups (Cyr, 2019). In the traditional setting, the discussion group must remain small so that each participants has a chance to speak. This small-N constraint limits the ability of synchronous group results to generalize inferentially to a larger population and to produce meaningful or valid discursive data.[3]

We use digital twins LLM simulation methods (Rountree et al., 2026) to explore the properties and behaviors of the structured group under alternative design choices. The simulation shows, among other things, that the small-N design often neglects unpopular viewpoints, and that among the viewpoints discussed, the qualitative text data collected is poorly correlated across replications, indicating that the small-N design does not systematically capture opinions within a population. In contrast, the large-N design robustly surfaces viewpoints supported by even small minorities of participants, and inferentially captures the conversation space that exists in a larger population. The structured group design is motivated by normative deliberative theory (Gutmann and Thompson, 1996) and contributes to an emerging literature on using LLMs to develop deliberative systems and platforms (Estornell and Liu, 2024; Friess and Eilders, 2015; Karacapilidis et al., 2024; Ma et al., 2024; Rountree et al., 2026; Tessler et al., 2024; Zhang et al., 2024).

The software to implement the structured group design – with either human or LLM-

[3]Of course, the structured group design is not appropriate if the opinion-formation hypothesis involves face-to-face interaction, such as the study of conflict aversion or emotions in discourse.

simulated participants – is open source, easy to use, and adaptable to any research topic, covariate set, or question. And more generally, the LLM pipeline we develop to evaluate synchronous designs generalizes to any set of design choices, topic of interest, and target population. We identify the assumptions required in order for the digital twins simulation to contribute inferentially toward the accumulation of knowledge regarding discursive designs, and we show validation methods that can help to weaken those assumptions.

2 The Structured Group Design

The synchronous group design enables the researcher to collect more nuanced data on considered opinions than can be captured in a standard survey. The group-based discussion replicates the kinds of everyday social interactions that generate opinions and can reveal meanings and understandings from the participant's own (emic) perspective (Cyr, 2019; Habermas, 1984), revealing not just what opinions and beliefs individuals hold but why they hold them and how they are formed (Chambers, 2018; Fishkin, 2018).

We posit that discovering the systematic discussion on a topic or issue within a population, or that would occur within a population, is a core goal of synchronous group research. As Fishkin (2018) notes, the synchronous group design should create a data generating process where the discussion data that results from the group matches or reflects the discussion that would occur within a larger population, had the larger population engaged in a similar discursive effort. Unfortunately, the traditional design for the synchronous group is unlikely to yield data that recovers this population-level perspective. In the traditional design, the number of participants must remain small – generally between six and eight participants per session – so that all participants have an opportunity to speak (Cyr, 2019). The small-N design creates vulnerabilities, however, that make the results likely not generalizable or representative of an emic, interpersonal understanding among members of a population for the issue under discussion (see Neblo et al., 2019).

We propose an LLM-assisted research design for a synchronous meeting that can scale to any N, which we call the *structured group* design. The user interface design and algorithmic curation pipeline to implement the structured group design for human participants has been implemented in the collabinar platform called *Prytaneum*,[4] a platform that is open source, web-based and free to use.[5] Scaling the synchronous group to a large N enables inferential research under relatively weak assumptions that we state below, and so addresses many of the limitations with the traditional, small-N synchronous group design. These assumptions can be further weakened with validation methods.

The structured group design involves three stages, for each question of interest. In the first stage, the facilitator provides background information and framing, the ground rules and motivation, and then presents a question to the audience with a set of closed-ended response options paired with a text box requiring participants to explain the reasoning why they chose their preferred option. For example, in our primary simulation below, we use the question, "What is your preferred approach to developing nuclear power?" With response options 1: "Implement rapid expansion and investment of nuclear power;" 2: "Maintain the current nuclear energy operations without change;" or 3: "Phase out existing nuclear plants and halt new construction." Of course, the question and standpoint options can be anything the researcher chooses. Once the facilitator closes the poll, the collabinar algorithm generates a summary of the participants' open-ended reasoning to support each of the four options. If none of the respondents chooses a given option there will be nothing to summarize and the summary for that option will be empty.

In the second stage, the facilitator shares the summarized reasonings with the structured group participants, and then asks the participants to read a reason supporting one of the standpoint options they oppose (among the options that have a non-empty sum-

[4]https://prytaneum.io/

[5]The structured group design can be implemented as an asynchronous survey using three waves. However, the asynchronous survey design would lack a facilitator and discussion lead, and would likely experience significant attrition across the three waves, and suffer from the limitations of asynchronous platforms such as reduced spontaneity, shorter answers, and uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

mary), and then to write a counterargument to that reasoning.[6] Once the second stage ends the algorithm summarizes the counterarguments submitted for each of the non-empty options. Then in the third stage, the facilitator asks participants to read the summarized counterarguments to the preferred option they selected in stage 1 and then asks them to write a response to the counterargument. The third stage responses are then summarized by the algorithm. Each of the summaries, along with all of the individual responses and all measured covariates, are stored in a database for post-session analysis.

The structured group design scales to any number of participants. Irrespective of the number of other participants, each participant has a robust opportunity to express their views regarding the question under discussion. Assuming a large number of participants, and provided the options are well-constructed, there should be sufficient reasoning to support each of the options to reflect the full distribution of perspectives in the underlying population of interest from which the participants are drawn, analogous to the concept of inclusive discursive representation in deliberation theory (Dryzek, 1990). Further, with a large N, the discussion summaries should inferentially capture systematic features of the underlying discussion, and hence should be stable over repeated experiments. In contrast, small-N groups under the structured design can yield final summaries that are inconsistent across experiments or exclude reasoning to support the options favored by unpopular or underrepresented perspectives.

We use digital twins simulation to explore the properties of the structured group under different design choices, including sample size, response skew, and stratified and simple random sampling. The full diversity and nuances of individual perspectives are captured in the responses across three stages of discussion that are stored in the database and can be mined for further insights.

[6] Alternatively, the researcher can randomly assign an opposing viewpoint in the second stage. A complete randomization ensures that each non-empty option will receive counterarguments.

3 Digital Twin Simulation Methods

We propose a general method to simulate the results of synchronous human interaction in order to understand the properties of alternative research designs. Specifically, we use LLM prompting to simulate the discursive data generation process using alternative design choices for the structured group design. These simulations allow us to explore the properties of different sample sizes, under differing sampling methods and response options, in repeated trials, at considerably lower cost and more efficiently than if we recruited humans to generate the data.[7]

To improve the simulation methods, we make use of a digital twins approach. Digital twins technology uses data from the physical world in order to anchor the simulation to the physical process being modeled, provided the simulation undergoes sufficient validation (National Academy of Sciences, 2024). There is considerable work on digital twin silicon sampling for survey research (Arora et al., 2025; Brand et al., 2023; Chuang et al., 2024; Dillion et al., 2023; Hämäläinen et al., 2023; Korst et al., 2025; Sarstedt et al., 2024), along with several projects that curate databases to power a digital twins pipeline (Hu et al., 2025; Park et al., 2024; Toubia et al., 2025). To create personas, we pipe digital twin covariate data into the LLM prompt, generating instructions for the persona the LLM is to adopt, as well as background information and the previous stages of the discussion as context which improves consistency of LLM performance (Zhang et al., 2024).

We created simulated structured group data by using a digital twins approach to design an ensemble of LLM personas – that is, a set of LLMs prompted to have diverse views driven by covariates that reflect the U.S. population. Here we make use of the digital twins survey database provided by Toubia et al. (2025). In the simulations, we sample from the 2,058 Toubia survey respondents and use the covariate information to create a unique digital persona via prompt engineering, requesting the LLM to adopt the persona reflected in the digital twin covariate values in order to provide open- and closed-

[7] The simulations in this paper cost about \$80 total and require about four days to run.

ended survey responses. Since the respondents in the Toubia digital twin database are representative of the U.S population, random samples from the database provide covariate information that reflects the joint distribution in the U.S. population.

Persona and response generation. The question and closed-ended standpoint option choices for the structured group design discussion, as well as the covariate set, are selected by the researcher. For example, below we focus on the topics of nuclear power, gun control and increasing the size of the U.S. House of Representatives. We specify covariates that explain variation in attitudes toward the topic to determine the digital twin "persona" of the LLM participant, along with an additional parameter that decides which LLM to use for that persona.[8] For example, in our simulation on nuclear power, the covariates are *education-level* (less than college degree, college or more), *gender* (male, female), *political ideology* (liberal, conservative), and *environmentalism* (pro-environmental movement, not pro-environmental movement). The prompt for nuclear power for stage 1 is:

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You must select from one of the three possible standpoints on nuclear power and briefly provide your reasoning for doing so in 1-2 sentences.[9] The possible standpoints are:
>
> - 1: Implement rapid expansion and investment of nuclear power.
> - 2: Maintain the current nuclear energy operations without change.
> - 3: Phase out existing nuclear plants and halt new construction.

[8]In the core simulation, we diversify the LLMs so the results are not confined to a specific LLM. We used 4 LLMs: Gemini 3.0 flash preview, Qwen3 32B, Llama 3.3 70B versatile, and OpenAI GPT OSS 120B, with the latter three being open weight LLMs. Each persona used the same LLM model and covariate values across the three rounds of the focus group.

[9]The prompt also provides background information as context to help the LLM model personalities. The background we use is taken from this PEW report: https://www.pewresearch.org/short-reads/2025/10/16/support-for-expanding-nuclear-power-is-up-in-both-parties-since-2020/. "To help you choose an option, please know that according to PEW Research Center, about half of US residents favor expanding nuclear power, while the other half prefer to maintain operations without change or to phase it out. Most men favor expanding it while most women prefer phasing it out or keeping it the same. Republicans are more likely to favor expanding compared Democrats. Environmentalists have mixed opinions, where some favor expanding it because nuclear has a low-carbon footprint, while others favor contracting it because they are concerned about the risks of nuclear waste disposal."

After the personas respond, we request Google Gemini to summarize the reasons offered for each standpoint option (among non-empty options) using the following prompt:

> You are a helpful assistant tasked with summarizing argumentative statements on the topic of nuclear power. I have gathered a list of statements that participants in an online focus group platform have provided as their reasoning to [standpoint option text]. Summarize the following statements as no more than five concise summaries. Your summaries should be worded as passive tone viewpoints, so that they capture as many viewpoints presented in the given statements as possible.

The LLM's responses in stage 1 create three inputs to the second stage: *stage1-standpoint* (the participant's chosen standpoint from stage 1), *stage1-reasoning* (the participant's reasoning for choosing stage1-standpoint), and *opposing-viewpoint* (a randomly selected summarized reason to support a standpoint that the persona did not choose in stage 1). The prompt for stage 2 is:

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: '[stage1-reasoning]'. As part of the focus group discussion, a different viewpoint has been brought up: '[opposing-viewpoint]'. You must now provide a counterargument for this viewpoint on nuclear power in 1-2 sentences.

The responses from stage 2 are summarized in a manner similar to the summarizations in stage 1, which creates one more input to the third stage, *counterargument* (a randomly selected summarized response from stage 2 that argues against this participant's original standpoint). The prompt for stage 3 is:

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: '[stage1-reasoning]'. As part of the focus group discussion, a counterargument to your standpoint has been brought up by a participant who said that: '[counterargument]'. You must now respond to this counterargument in 1-2 sentences.

The responses in stage 3 are summarized again within each (non-empty) standpoint. The simulation then saves the persona-level responses for each stage along with all inputs such as the covariates defining the persona, the opposing standpoint used in stage 2, and the counterargument used in stage 3. It also saves the summaries from each round. For illustration, the appendix provides the summarized responses for a single N=60 group on nuclear power, showing the arguments raised in the LLM-simulated structured group.

Simulating the sampling distribution for each design. To simulate the sampling distribution of the text data generating process, we have the LLM personas engage repeatedly in trials under each alternative design option, with different topics and survey prompts. Our focus is on assessing the consistency and stability of the summarized responses regarding each standpoint option across trials. To evaluate the consistency of text data across trials, we calculated pairwise cosine similarities for the text generated within each of the response summaries, for each standpoint option, across repeated simulated structured group trials, under each design (Schuld et al., 2024).[10] Cosine similarity is an appropriate comparison metric for our application since we are only making comparisons within a topic and within a stance or standpoint. This design avoids the well-known problem that, for example, inserting the word "not" in an argument can make two texts have opposing meaning but very high cosine similarity.

The cosine similarity of the summaries between two trials evaluates the stability of the summarized reasoning for each option expressed across the trials. If a synchronous design inferentially explores the full conversation space for a topic, then the summaries across trials will be semantically similar resulting in high cosine similarity scores on average with low variation across trials. If the design results in idiosyncratic discussions then the

[10]Cosine similarity is a measure of the semantic similarity between two strings of text that is widely used in natural language processing. To calculate the cosine similarities, we first convert the text output to embedding vectors which represent the semantic meaning of the relevant comparison text. We use the Hugging Face sentence transformer `all-MiniLM-L6-v2` to map each text response into a 384 dimension vector, and then apply the cosine rule. Cosine similarity scores range from zero (orthogonal text) to one (identical text).

summaries across trials will be dissimilar, resulting in low cosine similarity scores across trials and high variance. However, stability across trials by itself does not demonstrate an inference to a population-level discussion. This inference requires stability to be combined with two assumptions, coverage and unbiasedness, that we discuss below.

Running the simulator. The project repository houses all of the simulated output across all trials we report. A separate repository distributes the Python scripts needed to run the structured group simulator; one can use the scripts to replicate the findings of this paper, or to run simulations on any topic, covariate set, or question of interest.

4 Results

We conducted sets of digital twins simulations on the topics of nuclear power, gun control, and increasing the size of the U.S. House of Representatives. The nuclear power simulations use simple random sampling of personas to establish the random baseline sampling distribution for the structured group design under alternative design choices. Researchers who use the traditional, small-N synchronous design often use stratified sampling in order to ensure perspective diversity within discussion groups. To explore the consequences of stratified sampling, we use the highly polarized topic of gun control and stratify groups by liberal-conservative ideology. Finally, we use the human data on expanding the House of Representatives in order to conduct validation tests and replications.

Results under simple random sampling. We start with the nuclear power issue and use simulation to discover the sampling distribution of the structured group design under simple random sampling. We use the nuclear power question, three standpoint options, and prompts that we describe above. Making use of the structured group design, we ran 50 trials of the small-N version with six participants each; 30 trials of the medium-N version with 60 participants each; and 20 trials of the large-N version with 600 participants each. Each "participant" was one of the randomly selected digital twin LLM personas. Each trial produces a stage 3 summary text capturing the reasons participants offered to support and counter each standpoint.

Then for each standpoint, we create an $N \times N$ matrix of pairwise cosine similarities, measuring the similarity of the summary of stage 3 responses for each trial with the summary of stage 3 responses for every other trial within the standpoint and within that design. We retain only the lower triangle of this matrix, excluding the diagonal. If a summary is empty for a given standpoint on a given trial, the cosine similarity with every other trial is zero by definition.[11] For the $N = 6$ participant version trials we have 1,125 pairwise comparisons; for the $N = 60$ trials we have 435 comparisons; for the $N = 600$ trials we have 190 comparisons.[12]

Comparing these similarity scores across our nuclear power trials shows the relative stability of responses within a given structured group design under simple random sampling, within each standpoint option. Figure 1 compares the densities of the across-trial cosine similarity scores for designs with six, 60 and 600 personas, conditional on the response option. The within-response distributions of the cosine similarity scores show that the small-N version with size six has a mode at zero for two of the three standpoints, indicating that in many trials, no persona advocated these standpoint options (the summary was empty) with some frequency.

As a result, for groups with only six participants, that meant that for many trials there were no participants that favored one or more of the other options, and hence these other

[11]The empty text cells create a deterministic dependence in the pairwise comparisons. We account for those dependencies in two ways. First, in displaying the cosine similarity distributions in the main text, we construct histograms that visualize the mixture distribution, where cosine similarity is either zero or non-zero, and if it is non-zero we display the marginal distribution. Second, in the appendix we show two-dimensional scaling plots that give zero weight to the empty cells, and then show the pattern of dependence among the non-zero cells.

[12]Each 600 participant simulation takes about an hour to run. Fortunately, since the results are nearly identical across trials, we do not need to run many trials to recover the large-N sampling distribution and 190 is a sufficient number. We show in the appendix two-dimensional scaling plots that there is no dependence or clustering across trials in the large-N case and hence the 190 draws explores the full sampling space. Further, the appendix reports a complete replication using new LLM models, and the results for the large-N processes are nearly exactly identical between the original and replicated results.

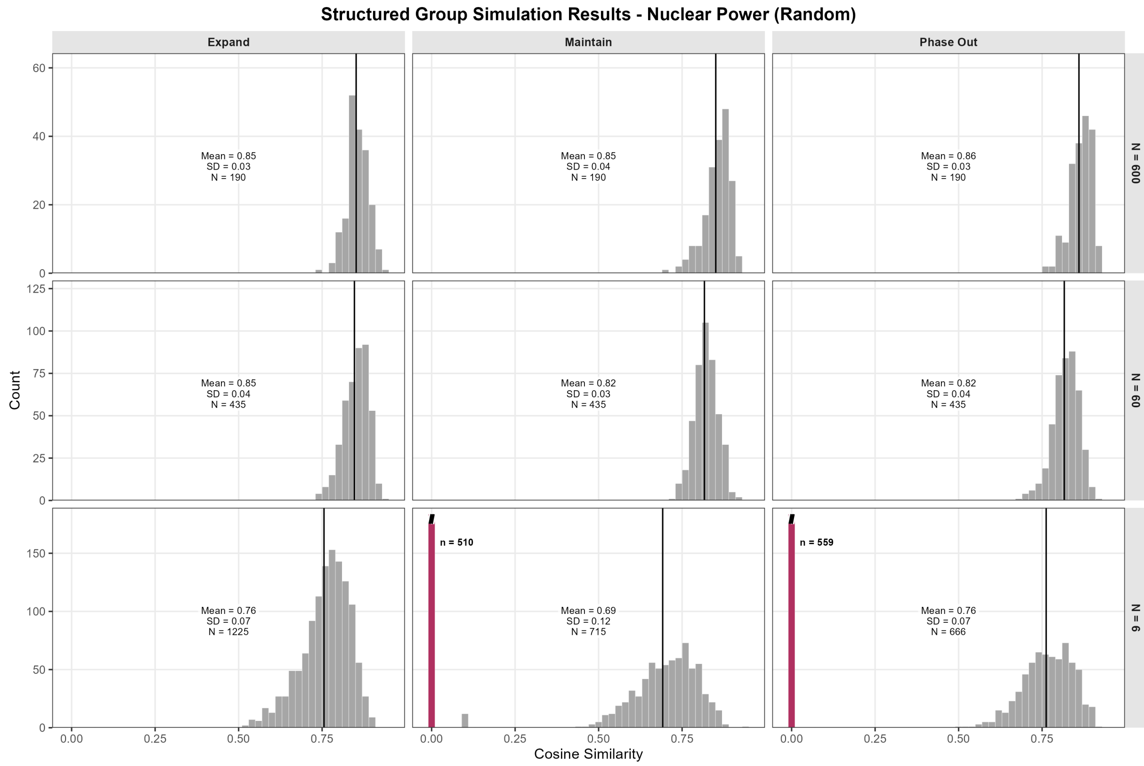

Figure 1: Distribution of pairwise cosine similarity scores, comparing between the $N = 6$ (50 trials), $N = 60$ (30 trials), and $N = 600$ (20 trials) group designs. Each panel visualizes a mixture distribution, where the cosine similarity score is either zero (dark bar) or non-zero, and if it is non-zero the marginal distribution is shown (grey bars). Simulated focus group results on the topic of expanding nuclear power – three response options. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction.

options were never even discussed in the structured group. In addition, when an option is discussed in a small-N group design, the cosine similarities indicate wide instability in the response across trials with non-zero modes centered at about 0.7 and wide dispersion – showing an interaction under the small-N design between the (in)frequency that an option is discussed and the idiosyncrasy of the discussion.

By comparison, the 600 participant focus groups have a unimodal distribution with relatively low variance across trials centered at about 0.9. Conceptually, one can think of the sequence of experiments in the small-N design is one that results in a multimodal distribution of divergent results while the sequence of experiments in the large-N design remains a stable, unimodal stationary distribution. And importantly, the degree of instability of the small-N design itself varies across the response options, while the large-N design returns results that are always stable and of similar magnitude across the response options, irrespective of the relative popularity of each option. The performance of the trials with 60 participants is between these two extremes.

This simulation shows limitations in the traditional small-N group design, and the merits of the large-N design. Conceptually, just as with quantitative research, we see a strong instability of results across trials for the small-N design. This means that the results for the small-N design are not inferentially reliable representations of the underlying discussion that is often the object of discovery in synchronous group research. Instead, small-N group dynamics create qualitative data that either reflects idiosyncratic discussion, or excludes unpopular viewpoints in the population. In contrast, the large-N design yields a draw from a stationary distribution that reflects a systematic summary of the discussion that does or could occur in the population. We can take these summaries from the large-N design as inferential to a population-level discourse if this stability is combined with two assumptions, coverage and unbiasedness, which we discuss below.

We explore the properties of additional design features under simple random sampling using two follow up experiments that we report in the appendix. To motivate the first follow up experiment, we note that there is both a normative reason and a safety reason to show participants a summary of the participant reasoning after stage 1 and stage 2. The normative reason is that, in showing participants the essential reasoning to support each option, participants are able to see the range of reasons supporting options they disagree with. The safety reason is that, absent strong filtering, the individual responses potentially expose participants to off-topic or inappropriate or harmful comments. At the same time, however, there is some information loss by summarizing the intermediate discussion stages. Figure A2 shows the degree of information loss by reporting slightly improved results from an alternative structured group design, where participants are shown randomly drawn individual responses instead of summarized responses. This experiment shows that there is a minor tradeoff between the normatively- and safety-motivated structured group design and the amount of information contained in the final summaries.

The second follow up experiment explores the effect of a heavily skewed response distribution. To create the skewed response distribution, we take advantage of the well-known property of LLMs to choose options that are least controversial. To this end, we add a fourth response option "Prioritize researching improved implementations of nuclear power," and as expected each of the LLMs choose this option nearly always, which makes the other options much less likely to be chosen. The results of this experiment are reported in Figure A1. Note that when the response distribution is relatively skewed, the performance of the 60 participant design nearly matches that of the 600 person design, but the performance of the six person design worsens. This indicates that a larger sample size is required in the presence of viewpoints supported by small minorities of participants, and improved instrument design can reduce the N needed for reliable results. This simulation approach to understanding the sampling distribution of the data is analogous to recent proposals to use simulation for power analysis under complex research designs (Blair et al., 2019). These results also show a possible tradeoff in discursive research, where placing constraints on the discussion can increase the stability of the results.

Results under stratified sampling. Given that a small sample of six or eight participants will have a considerable amount of sampling variability, synchronous group researchers understand that stratified or even purposeful sampling can improve on simple random sampling to ensure diversity in a given group (Cyr, 2019). For example, researchers interested in methods of ideological depolarization will compose small groups that balance liberals and conservatives (Esterling et al., 2021). We are able to use our digital twin simulation methods to explore benefits to small group research that can occur under such stratified sampling.

Of course, stratification does not help when covariates only weakly explain positions on topics such as nuclear power, but it might help improve the balanced representation of perspectives for topics where the stratifying covariates predict responses. Here we explore the consequences of stratification on self-reported ideology on the issue of gun control.[13] We build the LLM personas using the same gender, education, ideology and model covariates we describe above, and run trials for $N = 6$ where we stratify sampling to ensure that half of the participants are liberal and half are conservative, $N = 6$ with simple random sampling, and $N = 60$ with simple random sampling. This setup allows us to compare the stratified sampling design to the two random baselines.

We set three standpoint options for the LLM personas to choose from: "Gun control laws should be more strict," "Gun control laws are about right," and "Gun control laws should be less strict." For context, we include the background information text: "To help you choose an option, please know that according to the PEW Research Center, 82 percent of U.S. conservatives believe that gun control laws should be the same or less strict while 92 percent of U.S. liberals favor making gun control laws more strict. That is, liberals favor making gun control laws more strict and conservatives to keep the laws the same or make them less strict."

Figure 2 shows the results. Compared to the small-N simple random sample design, stratification by ideology does help to ensure balance across the two polarized response options, to either be more restrictive or less restrictive, but it has no effect on the more neutral option to advocate that gun control laws are currently the right amount of regulation. The results show that this stratification design is effective if the purpose of the discussion is to ensure participants are speaking across differences, but less effective if the purpose of the discussion is to inferentially discover the full underlying discussion space (again, under the coverage and unbiasedness assumptions we discuss next).

[13]For background, see the PEW report https://www.pewresearch.org/short-reads/2024/07/24/key-facts-about-americans-and-guns/

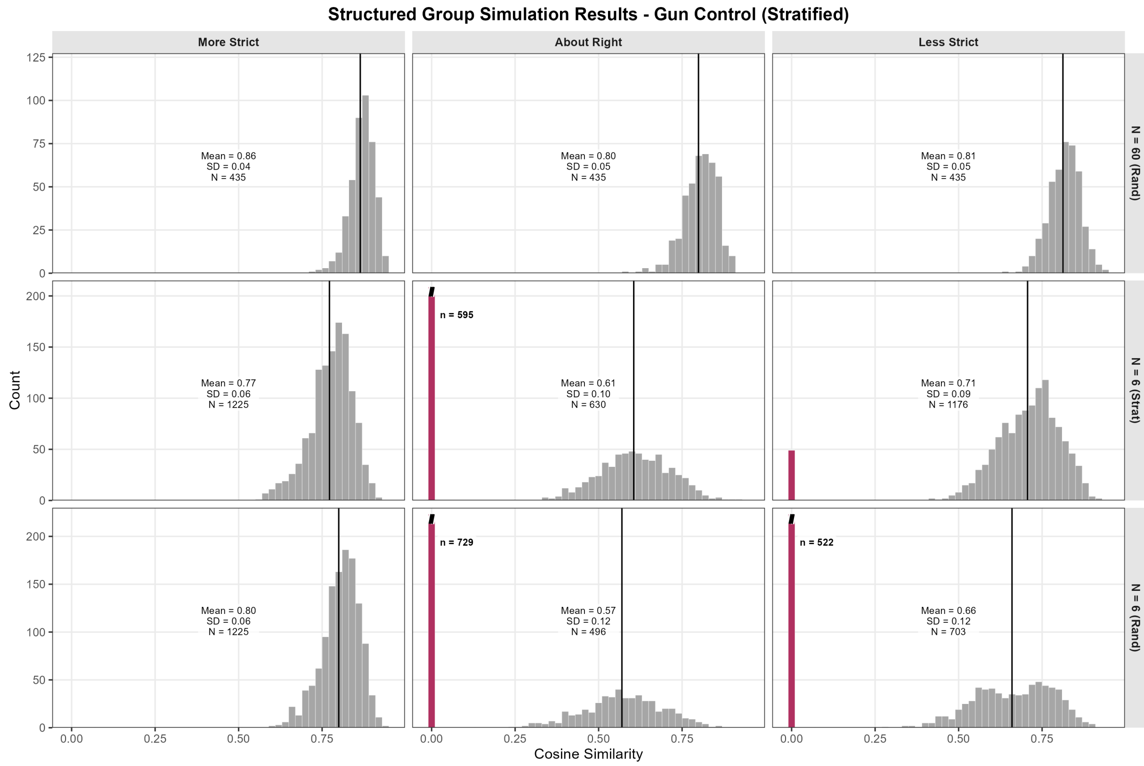

Figure 2: Distribution of pairwise cosine similarity scores, comparing between the $N = 6$ (simple random sampling, 50 trials), $N = 6$ (stratified sampling, 50 trials), and $N = 60$ (simple random sampling, 30 trials) group designs. Each panel visualizes a mixture distribution, where the cosine similarity score is either zero (dark bar) or non-zero, and if it is non-zero the marginal distribution is shown (grey bars). Simulated focus group results on the topic of gun control – three response options with stratified sampling comparison to two random baselines. Response 1: Gun control laws should be more strict. Response 2: Gun control laws are about right. Response 3: Gun control laws should be less strict.

5 Assumptions and Validation

As with all LLM-based technologies, digital twin methods have known limitations that can reduce their ability to add to the accumulation of knowledge (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). In addition to hallucinations, LLMs are known to exhibit systemic biases inherited from the training data (Blodgett et al., 2020; Borkan et al., 2019; Caliskan et al., 2017; Feng et al., 2023; Hofmann et al., 2024; Hu et al., 2024; Motoki et al., 2024; Sorensen et al., 2024; Taubenfeld et al., 2024; Tjuatja et al., 2024) or from alignment processes

(Santurkar et al., 2023). Digital twins and silicon sampling methods can introduce other distortions. For example, when prompted to respond from the perspective of a specific persona, LLMs can flatten or misrepresent the perspectives of identity groups (Sarstedt et al., 2024; Wang et al., 2025). LLMs also perform less well in representing non-English speaking populations (Qu and Wang, 2024).[14]

As a result, the extent to which digital twin simulation can be relied on to add to the accumulation of knowledge depends on a set of application-specific assumptions (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). While it is common for researchers to articulate assumptions for validity in statistical applications (Esterling et al., 2025), to date the LLM literature has not developed norms to state assumptions required for simulation research.

Our application of digital twins simulation to evaluate synchronous group design relies on two core assumptions: 1) *Coverage:* with enough participants, the LLM-based discussion recovers the full range of human-made arguments within a target population on the topic of discussion across the three stages, or that could be made, and 2) *Unbiasedness:* that the discussion summaries do not systematically exclude the perspectives of identity groups. Given these assumptions, we can draw inferences regarding the discussion space in the underlying population. These are relatively weak assumptions that can be further weakened by our validation tests.

Importantly, note that there are two further, and relatively strong, assumptions that would be required in the ordinary case of survey research but that we do not need. These are 3) *Faithfulness:* that the digital twin personalities produce choices, reasoning or content that reflects the (intersectional) demographics of the digital twin covariates. It is well known that the documents in the LLM training text data are not associated with demographic covariates, and hence LLMs are not well-designed to have the LLM

[14]In principle, LLM output can be debiased using regularization, fine-tuning, or alignment processes such as reinforcement learning from human feedback (Agiza et al., 2024; Bang et al., 2021; Bolukbasi et al., 2016; Dixon et al., 2018; Jin et al., 2021; Lin et al., 2024; Liu et al., 2020, 2021). Debiasing would be difficult or impossible to implement with human participants.

output match the voice of identity groups defined by covariates (Wang et al., 2025). In our application, we use digital twins prompting to ensure that the arguments that LLMs voice in the group discussion ecologically capture the diversity of human argumentation, but our application does not require the argument content at the persona level to correspond to the demographic identity group's voice reflected in the covariates.

We also do not require the assumption 4) *Distribution match:* that the digital twin personalities select the standpoint options in a way that matches the marginal distribution of those preferences in the target human population (Bisbee et al., 2024), either conditioned or unconditioned on covariates. Our structured group design is motivated by the notion of discursive representation (Dryzek, 1990), and hence we only need to have a viewpoint expressed by some LLM personas for that perspective to be included in the corresponding summary. The marginal distribution of preferences among the participants within the discussion is not strongly relevant to discursive representation.[15]

We rely on validation tests to weaken our two required assumptions, coverage and unbiasedness. The first test weakens the *coverage* assumption. In this test, we compare the generated responses among the LLMs to a known list of arguments that have been generated by humans on the corresponding topic, and evaluate whether the LLMs express the full diversity of human viewpoints on the topic. Here we make use of an existing reference listing of pro and con arguments regarding nuclear power from the *Encyclopedia Britannica*,[16] which provides three pro arguments that nuclear power: 1) has zero emissions and can reliably and inexpensively support an electricity grid; 2) is a perfect complement to weather-dependent renewable energies; and 3) protects American global national security interests; along with three con arguments that nuclear power: 4) is dangerous, from the risk of meltdowns to the problem of nuclear waste; 5) is expensive to initiate and delays implementation of truly sustainable and renewable energy sources; and 6) is inextricably

[15] For example, if 100 LLM personas offer an identical perspective, and only one LLM persona expressed a different perspective, then the summary would include both perspectives. The marginal distribution of participants holding a given perspective is not strongly relevant.

[16] https://www.britannica.com/procon/nuclear-power-debate

linked to lethal nuclear weapons.

We scale this test by using Google Gemini to code the matches between individual arguments and the reference list. To confirm the accuracy of the LLM, we provided both a human coder and an LLM-based coder a candidate list of 60 persona-generated reasons for and against nuclear power and asked each coder to decide if each candidate reason matches with an argument on the Britannica reference list, and if so, which one. The two coders disagreed on 22 out of the 60 classifications. The lab team reviewed each instance of disagreement and found that seven disagreements were cases where the candidate reason expressed both reference arguments (so both coders correct), the human coder made 11 errors and the LLM coder made four errors.

We used the LLM to code classifications for a total of six trials of 60 persona participants. In each trial, the full set of reasons was given by the personas with the exception of the sixth (con) reason. However, it is reasonable to argue that the final con arguments, drawing the analogy to nuclear weapons, is a strange and not especially strong argument – so its absence from the discussion is not surprising in retrospect. In addition, the LLM personas expressed additional arguments that are not found on this reference list. Often, the diversity of arguments emerged over the course of the discussion. For example, in the first trial we coded, the first four of the *Britannica* arguments were expressed in the first round of discussion but the last two con arguments, arguments 5 and 6, were not. However, argument 5, that investing in nuclear power diverts funding from renewable sources, was raised repeatedly in the counterarguments and responses to counterarguments in stages two and three of the discussion. One can see all of the summarized arguments across the three stages for this example trial in the appendix. The Python script to run the LLM classification tool is available in the project code repository.

Next, we develop a bias audit method to detect bias in the summarizations (Altemeyer et al., 2025; Mittelstadt et al., 2023; Zhang et al., 2020). Note that the stable distribution that results from the large-N design is both normatively and inferentially defensible only if the qualitative summaries of the responses are an unbiased reflection of the individual contributions made during the meeting. If the AI-based summary tool is biased, then the summaries that result would be objectionable in that they would systematically exclude certain perspectives. It is well-known that the underlying LLMs that drive AI applications can have biases inherited from their training data or from alignment interventions (Borkan et al., 2019; Caliskan et al., 2017; Kambhatla et al., 2022; Liu et al., 2020; Santurkar et al., 2023). To test for *unbiasedness*, we use cosine similarity to measure the closeness of each persona's response to the corresponding summary (that is, the summary for the option they chose) and evaluate whether distances from the summary are random or if they are systematically related to the demographic covariate values (Wang et al., 2025). We conduct this audit using a human-generated dataset which we report in the next section.

6 Replication and Bias Audit with Human Data

To conduct a replication and to evaluate bias in the LLM summaries, we make use of a human dataset that we were provided that links open-ended response text to demographic covariates. The data come from an online Deliberative Town Hall (Neblo et al., 2018) held by the Institute for Democratic Engagement and Accountability (IDEA) at the Ohio State University.[17] The town hall discussion centered on topics related to the modernization of Congress, and the participants were a U.S. nationally-representative sample. After the town hall, the participants were asked to respond to a survey that collected demographic information and then they were asked to indicate their support for or opposition to a proposal to expand the size of the U.S. House of Representatives, and they also were asked to provide their reasoning for their position as an open-ended response. This data is very similar to the kind of data that would be collected in a structured group design in that the responses with the reasoning that participants submitted followed from a synchronous town hall discussion where they were exposed to both pros and cons.

[17]The data was collected under the Ohio State University IRB protocol 2019B0075.

There were 586 responses to the question on expanding the size of the U.S. House of Representatives.[18] First we use our LLM summary tool to create summaries of the favor and oppose human responses, and then we run our simulator with $N = 60$ on the same question in order to see if the LLM simulation replicates the human responses. We give the details regarding the persona covariates, question wording and context information in appendix B. In appendix tables A2-A5, we reproduce both the human-generated summaries and the LLM-generated summaries, and we label each unique perspective in order to evaluate the correspondence between the human responses and the LLM responses.

Among the human perspectives that disfavored expanding the House, the LLM personas replicated arguments that increasing the size would burden taxpayers, make Congress unwieldy, take the focus off of member accountability, and that the existing structure is adequate. The LLMs did not mention the human arguments that increasing the size might exacerbate gerrymandering or increase partisanship, and that new communication technology makes expansion unnecessary. Among the human perspectives that favored expanding the House, the LLM personas replicated arguments that smaller districts would improve local connections and access, increase diversity, improve legislative efficiencies, reduce the power of special interests, make leadership power less concentrated, and match the increasing complexity of the American population. The LLM personas did not mention the human argument that reducing district size would restrict gerrymandering.

Overall, the LLM personas give remarkable coverage of the human-generated perspectives. Among the seven human arguments in favor of expansion, the LLMs reproduced six (and offered one perspective not mentioned by humans). Among the seven human arguments opposed to expansion, the LLMs reproduced four. In retrospect, the arguments for and against expansion that referenced gerrymandering are not especially strong arguments, in that expansion and gerrymandering are almost certainly unrelated. And the argument against that referenced communication technology is likely unique to the

[18] We dropped responses with fewer than 10 characters.

IDEA town hall since that was hosted on a webinar platform. Finally, we note the LLM personas matched the human pattern, that we report next, that there are more diverse ways to oppose expansion than to favor expansion even among the human respondents.

Next, to evaluate bias towards identity groups in the LLM-based summaries, we calculate the cosine similarity of each (human) individual's reasoning with the summary of the position they chose. We ask the summary tool to include five different perspectives within each summary. We then calculate the cosine similarity between the individual's response and each of the five perspectives for their standpoint, and then retain the maximum value. That is, if individual i was in favor of expanding the size of the House, we calculated the cosine similarity between their response and the five in-favor summarized perspectives, and then saved the maximum of these five values.

Figure 3 shows the distribution of these individual-level cosine similarity scores, conditional on demographic covariates and on support or opposition position. Note that the cosine similarity is higher for the support than the opposition position, which simply indicates that there are more diverse ways to express opposition than support. But conditional on position, the distributions do not show strong substantive, systemic disparities across the demographic categories. The view of college-educated individuals who are in favor of the reform are slightly over-represented, which is not a surprise given that LLMs tend to be biased in favor of well educated responses (Santurkar et al., 2023).

Importantly, certain views of non-white respondents who are opposed to the reform seem to be slightly under-represented, given the mildly bimodal conditional distribution, which indicates that some of the perspectives in this group are not well-captured in the summaries. However, the bias audit helps to automate the detection of excluded perspectives since it is straightforward to surface the perspectives of those in the mode. We manually inspected the responses that composed this mode of the conditional distribution and found in each case, the semantic perspective was captured in the summary, but not the surface-level grammar of the expressed response such as in the example, "Too many

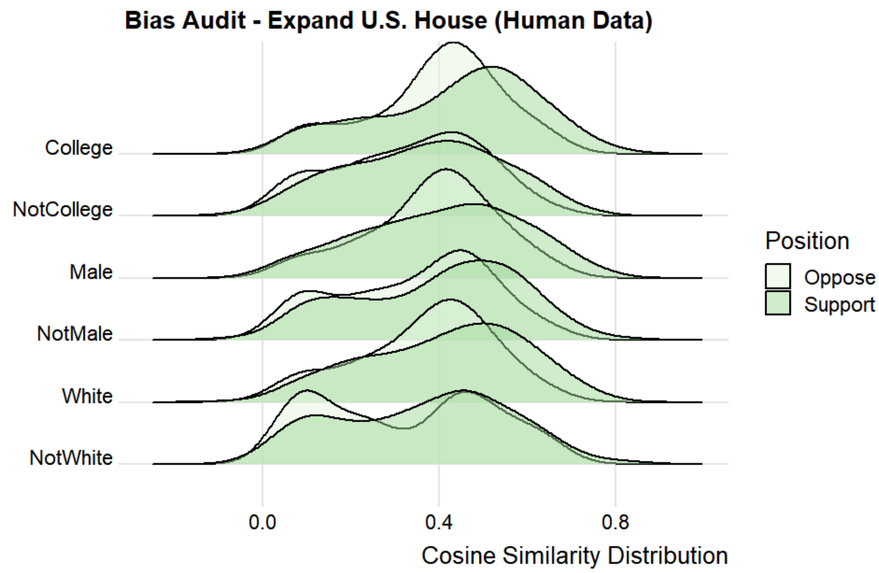

Figure 3: Distribution of cosine similarity scores, conditional on demographics and position. Here, cosine similarity is between 1) an individual's open-ended response and 2) the summary corresponding to the individual's chosen standpoint. Higher cosine similarity indicates the summary captures the individual's perspective, while lower cosine similarity indicates the summary excludes the individual's perspective. Topic: Expansion of the U.S. House of Representatives

cooks spoil the soup." And of course, also in contrast to the traditional synchronous group design, our method allows for this kind of bias audit to detect systemic disparities, and the AI-based summarizer can be debiased so that the summaries are orthogonal to demographic characteristics in the event that they do fail an audit (Bolukbasi et al., 2016; Dixon et al., 2018; Liu et al., 2020). In a traditional group design, where individuals speak in turn, there is no method to detect or to correct these kinds of systemic biases that occur in real time during a free-flowing discussion.

7 Qualitative Inference to Population-Level Discourse

The high variance of cosine similarity scores for the summaries across trials for the small-N design indicates that each discussion was in some sense idiosyncratic, and hence shows that the results can not be taken as generalizable inferences to a population-level discussion.

In contrast, the stability of the cosine similarity scores for the large-N design suggests that the summaries were capturing perspectives on the topic systematically. However, this stability by itself does not justify an inference to discourse in a larger population. To make this inference deductively requires the coverage assumption to ensure that all perspectives are included, and the unbiasedness assumption to ensure that the summary does not exclude perspectives. All scientific deductions rely on assumptions (Esterling et al., 2025), irrespective of the epistemology of the method. We propose coverage and unbiaseness as minimum assumptions necessary for the simulation to contribute to the accumulation of knowledge in synchronous design research.

8 Discussion and Conclusion

We propose an LLM-based digital twins pipeline that can be used to evaluate the properties of different synchronous designs. Digital twins simulation enables the researcher to explore the properties of alternative discussion designs computationally at low cost and effort by simulating the sampling distribution of qualitative data under the different designs. These methods enable the researcher to computationally understand the complex, emergent nature of human discussion under a given design, similar to how researchers use computational methods to understand the power of complex statistical models when an analytical solution is unavailable (Blair et al., 2019). These methods can be used to pilot alternative designs prior to investing in human trials. Finally, we state assumptions required for these simulations to add to the deductive accumulation of knowledge for discussion designs, as well as validation tests that can weaken these assumptions.

As a proof of concept, we apply our simulation pipeline to evaluate alternative designs for synchronous group research. The application of our methods using the structured group design helps to show the properties of designs for human interaction. All of these applications use digital twins simulation to demonstrate the clear superiority of our proposed large-N structured group design to that of the traditional small-N design. The traditional synchronous group design results in qualitative data and inferences that lack generalizability. In contrast, the large-N group design enables a large number of participants to, in effect, speak simultaneously and to understand all of the diverse perspectives among participants. We also show that stratification can help for cases where responses are well-predicted by covariates.

Further, we propose the coverage and unbiasedness assumptions needed to understand if the simulation can advance the accumulation of knowledge on research design, and validation tests designed to weaken those assumptions. In particular, we show that the LLM summary tool at the heart of our design is inclusive of diverse perspectives and does not seem to be vulnerable to systemic biases that can plague LLM-based tools. The stability of the summarized responses across trials, combined with the coverage and unbiasedness assumptions, allow us to make inferences to population-level discourse and the formation of public opinion.

Of course, the structured group design is not useful or appropriate if the focus of one's research is an investigation of deeply interpersonal human interactions, such as on the role of conflict aversion or emotions in small group settings. Further, the small-N design remains useful as a tool for exploratory qualitative research (Cyr, 2019). But there is no reason to limit synchronous group methodology to exploration. Our proposed simulation methods show how synchronous group designs can extend to inferential qualitative research, and can help researchers uncover a nuanced, interpersonal understanding of the merits of alternatives regarding a decision, in a way that meets the normative standards of deliberative theory (Dryzek et al., 2019; Niemeyer et al., 2024).

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Using Digital Twin Simulation for Synchronous Communication Research: With an Application to the Structured Group Design

Supplemental Information Appendix

A Validation Data and Supplemental Results

Validation data. To assess the ability of our LLM to match individual arguments to a reference list of arguments, we make use of a single trial for a focus group with $N = 60$, for a total of 180 human labels. After the research assistant completed the hand labeling, the entire lab team reviewed each of the 180 labels. In this trial, the LLM personas debated the question of regulating nuclear power that omitted the non-regulatory option to continue researching nuclear power. Since each trial is a random draw from the LLM, it does not matter which trial we investigate – so we simply used the first trial as our data source for the validation. To find the validation data, navigate to the reproduction GitHub repository to find the data in the folder with trial id 4260266019. Table A1 provides the text summaries for the three stages of the LLM persona discussion for this trial, where the first stage summarizes the initial reasons given, the second stage summarizes counterarguments to those reasons, and the third stage summarizes the reactions to the counterarguments.

Supplemental validation results. We supplement our main validation tests with a human verification to ensure the LLMs are correctly responding to the prompt instructions – that the LLM is adopting the instructed persona and responding appropriately to the initial survey question and to the other LLM inputs. To do this human verification, we make use of two tests, relying on the framework for "algorithmic fidelity" proposed by Argyle and her collaborators (2023). The first test is *backward continuity*, which requires that the LLM responses are consistent with the digital twin persona. To test backward continuity, we mask the digital twin persona instructions that were included in the prompt and ask a research assistant to infer the covariate values from the LLM generated text. The second test is *frontward continuity*, which tests whether the LLM output is a sensible response to the prompt. To test for frontward continuity, we ask a research assistant to read the prompt and assess whether the generated text is a natural or sensible response to the prompt.[19]

We first consider the backward continuity test, where we masked the persona covariate information and asked a research assistant to infer the covariate values based on the generated text alone for 60 observations on the topic of nuclear power. This test checks if the LLM is following the prompt instructions to create a given persona. We rely on χ^2 tests from confusion matrices. We find that the research assistant is 26 percent more likely to correctly infer female rather than male ($p = 0.063$), 31 percent more likely to correctly infer college educated rather than not college educated ($p = 0.022$) and 25 percent more likely to correctly infer liberal rather than conservative ($p = 0.036$). In addition, the true environmentalism score is about a standard deviation higher when the research assistant inferred the persona was positive toward environmentalism ($p < 0.001$). Finally, we also masked which standpoint option the LLM chose, and the research assistant was able to infer the standpoint chosen from the text of the reasoning offered and made the correct inference for 58 out of the 60 responses. Note that backward continuity weakens both the coverage and the faithfulness assumptions, although in our application we only need it to weaken the former.

Second, we consider the frontward continuity test, where we asked the research assistant to read the prompt created for each persona and to evaluate if the LLM output was responsive to the text in the prompt. This is similar to a Turing test. Here the research assistant was asked to read all three rounds for a 60 participant trial. Out of the 180 opportunities for the LLM to be non-responsive, the research assistant found only five cases where the LLM did not respond to the prompt. However, when the lab team conducted a review of those five responses, we found the output to be responsive in each case. By design, our pipeline does not depend on the memory of the LLM across the three rounds – instead, we pipe in the text from each previous round into the prompt. This method

[19] Frontward continuity stands as an informal Turing test (Aher et al., 2022), which requires that LLMs generate text indistinguishable from human generated text. A formal Turing test would randomize between LLM and human responses and ask the research assistant to guess which response is human.

avoids the limitations of LLMs' diminished performance in multi-turn conversations, and as a result the LLMs are responsive on each API call. One concern might be that LLMs are even more responsive to the prompt text than humans would be, creating a hyper responsiveness that would not be present in human-generated text.[20]

B Additional Simulation Results

Simulation under skewed responses. Figure A1 shows the results of the additional experiment that explores that consequences of heavily skewed response data on the stability of the qualitative results. Here we make use of the well-known behavior of LLMs to hedge when given the option. As we describe in the text, this experiment adds a fourth standpoint choice to recommend continuing research on nuclear power rather than make a regulatory decision such as expanding, maintaining or phasing out nuclear power. As Figure A1 shows, the LLMs chose the option to research nuclear power with the greatest frequency, which in turn increases the frequency that the other options were never selected – but only in the $N = 6$ design. Even with highly skewed response data, the $N = 60$ and $N = 600$ always had each perspective represented. The skewness degraded the performance of the $N = 6$ design considerably, and the $N = 60$ design slightly, compared to the $N = 600$ design. These results show that the simulation provides results that are similar to computational power analysis, where the researcher can decide if the slight decline in performance justifies a massive increase in sample size or not. It also shows that improved question design can reduce the number of needed participants, and the tradeoff that placing constraints on a discussion can improve stability.

Simulation with no summaries. Figure A2 shows the results of the additional experiment that uses where we show participants randomly selected individual arguments rather than summarized arguments. The structured group design shows participants sum-

[20] This is equivalent to filtering human response data to include only on-topic, substantive responses.

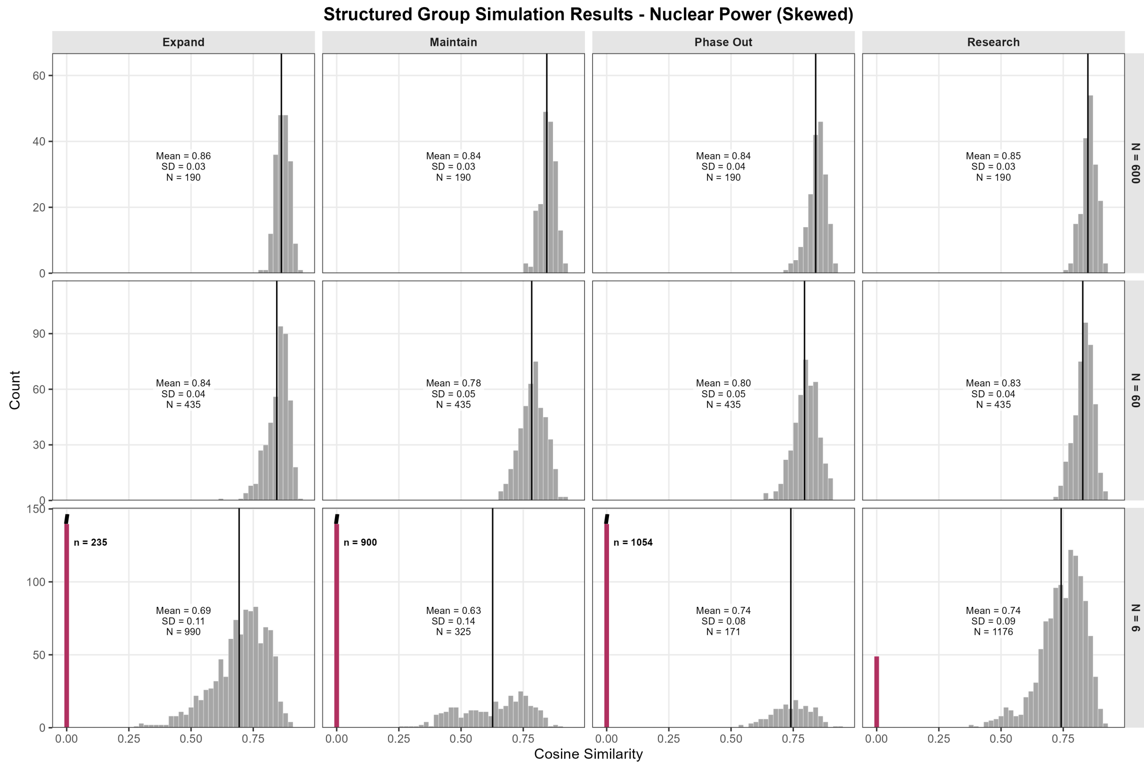

Figure A1: Distribution of pairwise cosine similarity scores for stage 3 summaries across trials, comparing between the small-N group design and the medium- and large-N design under random sampling. Simulated structured group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction. Response 4: Prioritize researching improved implementations of nuclear power.

marized arguments both for normative reasons – to ensure that each participant sees the full range of reasons to support or opposed each option – and for safety reasons – to ensure that participants are not exposed to off-topic or harmful content. This additional experiment demonstrates that using summaries instead of raw responses at stage 2 and stage 3 results in a loss of information. That is, the means are higher and standard deviations are lower for each of the designs in this alternative design. These results show that the normative and safety considerations result in a tradeoff between normative merit and the amount of information expressed.

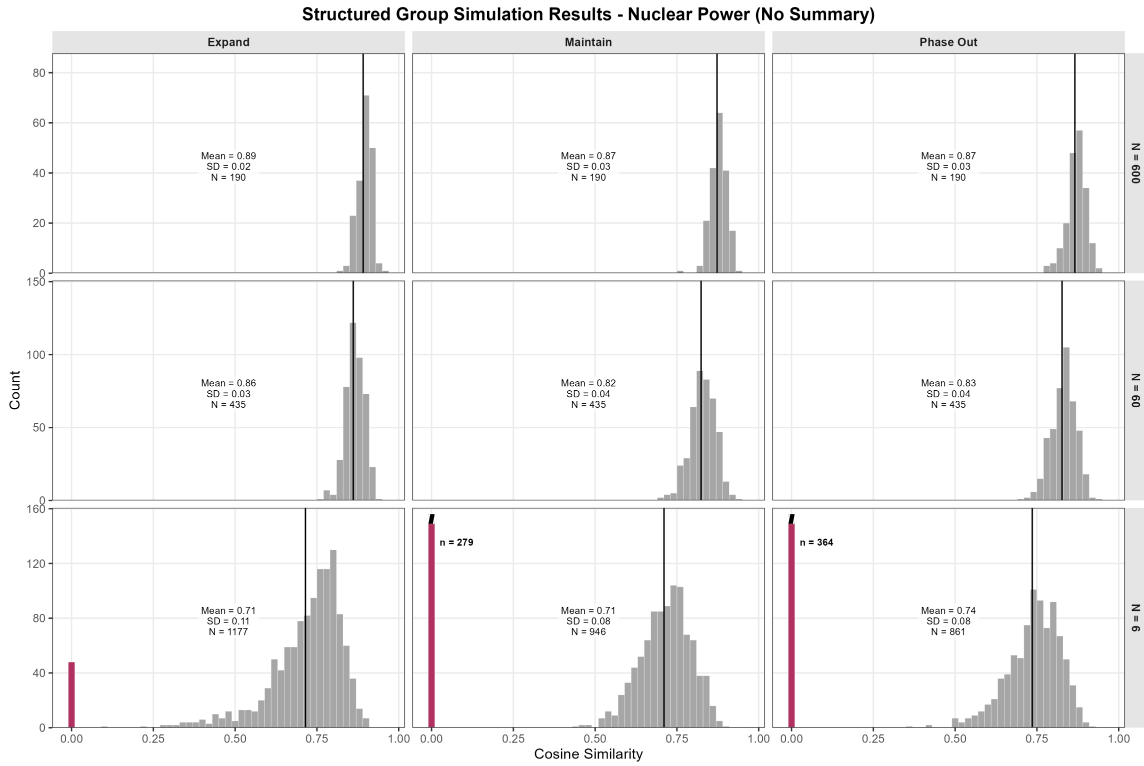

Figure A2: Distribution of pairwise cosine similarity scores for stage 3 summaries across trials, comparing between the small-N group design and the medium- and large-N design under random sampling. Simulated structured group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction.

MDS plots Figure A3 shows the smallest space two-dimensional representation of the cosine similarity scores for the simple random sample design for nuclear power, and A4 shows the same for the stratified sampling design for gun control. (The corresponding figures for the other experiments are available in the reproduction repository). These figures show the degree of dependence among trials under each design. Since empty text trials have a deterministic zero similarity, these are given zero weight in the plot. The plots show that there is some dependence in the $N = 6$ design but essentially no dependence in the $N = 60$ and $N = 600$.

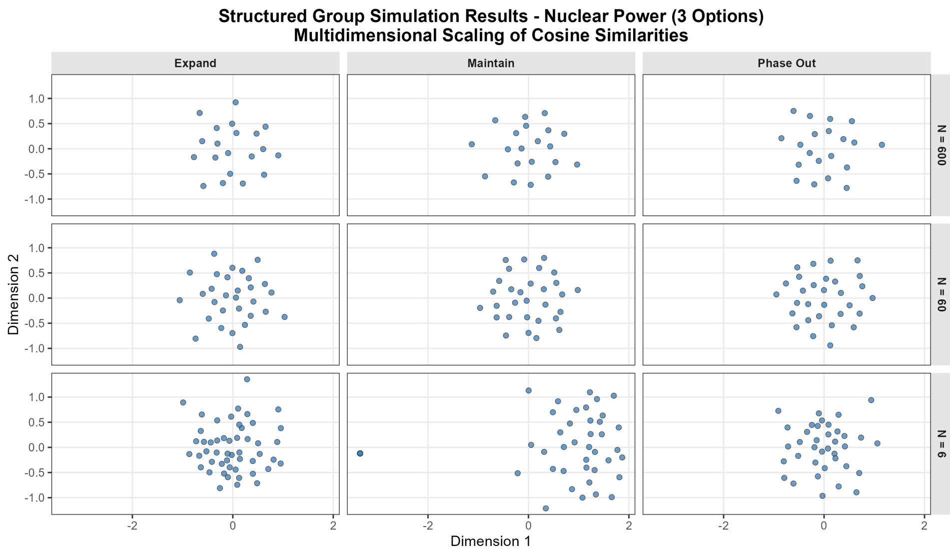

Figure A3: MDS plot of simulated structured group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction.

Simulated Replication of the Ohio State Deliberative Town Hall. For comparison to the human data results we report in the text, we replicate the Ohio State Deliberative Town Hall that focused on the topic of the expansion of the U.S. House of Representatives as a digital twins simulation using the structured group design. We implement the simulation with two response options: 1) increase the size of the U.S. House of Representatives by adding more seats, and 2) keep the size of the U.S. House of Representatives the same as it is now. To build the digital twin personas, we use the same pipeline and the same gender, education and ideology covariates as we used on our nuclear power simulation; we remove the level of environmental movement support covari-

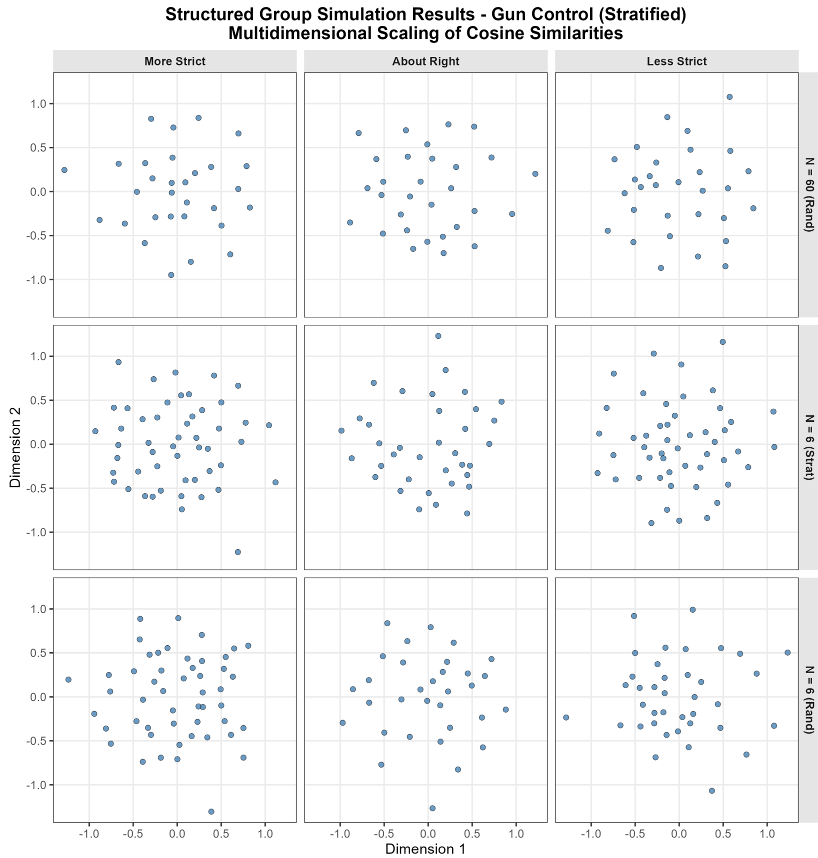

Figure A4: MDS plot of simulated structured group results on the topic of gun control. Response 1: Gun control laws should be more strict, Response 2: Gun control laws are about right, Response 3: Gun control laws should be less strict.

ate; and add a race covariate, a need for cognition score, and a vertical collectivism score which denotes a focus on group loyalty combined with acceptance of hierarchical structures.[21] As background information, we also include in the prompt "To help you choose an option, please know that expanding the size of the House will reduce the number of constituents in each district, enabling representatives to better connect with their district residents. However, adding new seats to the House would increase the cost of government and potentially make Congress more unwieldy."

[21]We add the need for cognition and vertical collectivism scores to the personas in order to ensure diverse responses to the prompt. In the absence of these covariates, liberal LLMs almost always chose to expand and conservative LLMs almost always chose to keep the House the same. The simulator repository provides a useful script to run stage 1 only of the experiment, which can be used to experiment to find the correct or optimal covariate set, such as if one wanted to meet the distribution match assumption.

Here we find that even with only two responses, there are many sessions where the small-N six person focus group never discusses the option to keep the House the same size. These are groups where there is no disagreement. And as before, when a topic is discussed, the discussion shows considerable variability in the cosine similarity scores, indicating that each event produces relatively idiosyncratic discussion and data.

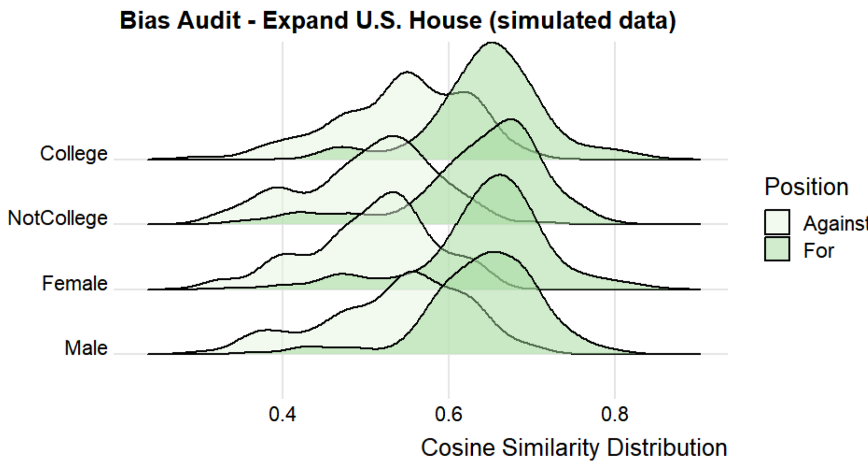

Figure A5: Distribution of simulated individual cosine similarity scores, conditional on demographics and position. N=600 design. Topic: Expansion of the U.S. House of Representatives

Figure A5 shows the bias audit results for the simulated data. Notice the strong similarities to the results of the bias audit between the data generated by humans as reported in figure 3 and in particular little evidence of bias in the summaries since the conditional distributions strongly overlap. The exception is that, as in the human data, college educated personas tended to have their responses more closely reflected in the summaries. Interestingly, even in the simulated data, supporters tend to have their arguments more closely reported in the summaries than opponents, which again suggests that there are more ways to oppose the policy than to support it. That the simulated data reflects these patterns helps to reinforce that the digital twin simulated arguments have good correspondence with human authored arguments.

Bias Audit of the Summaries of Simulated Nuclear Power Responses. Here for completeness we show the bias audit for our main simulation on the topic of nuclear power. To create the simulated audit data, we calculate the cosine similarity of each individual LLM-simulated persona's reasoning with the summary of the standpoint option they chose. Recall that we ask the summary tool to include five different perspectives within each summary. We calculate the cosine similarity between the simulated persona's response and each of the five summarized perspectives, and then retain the maximum value. That is, if simulated individual i chose response option k, we calculated the cosine similarity between i's response and the five response k summarized perspectives, and then saved the maximum of these five values. Here we audit the summaries based on the stage 1 discussion.

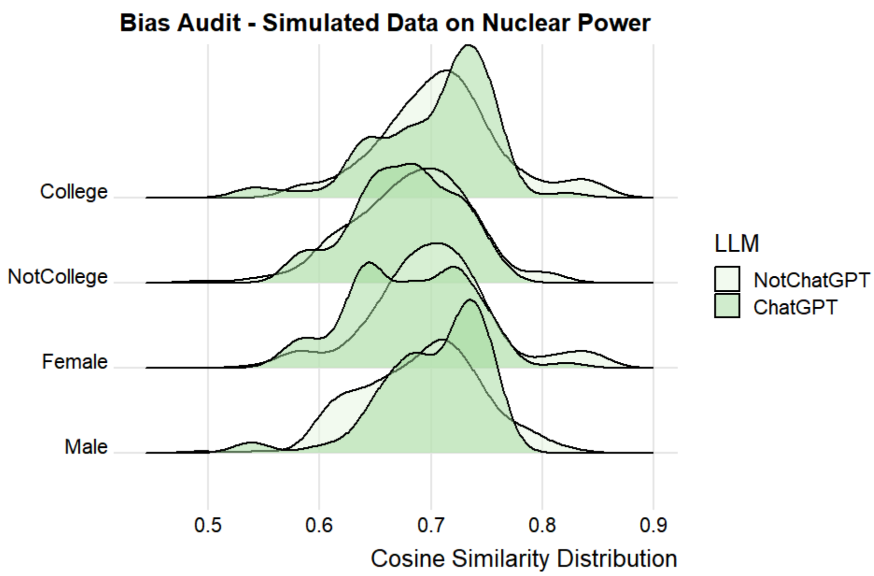

Figure A6: Distribution of simulated individual cosine similarity scores, conditional on demographics.

Figure A6 shows the conditional distributions of the cosine similarity scores for the simulated individuals, conditional on two covariates (education and gender) and the LLM (OSS-GPT versus the others). The conditional distributions indicate no systemic disparities by gender, or perhaps an under-representation of the female non-OSS-GPT perspectives. However, there is a slight over-representation of college-educated simulated individuals. As we show above, these results replicate in the human data. Finally, there is a noticeable over-representation of OpenAI's OSS-GPT, which is likely due to the fact that GPT provides more generic answers to the prompts.

Overall, in both the simulated and in the human-generated data, the conditional cosine similarity distributions in the bias tests are largely the same across all covariates. These results suggests that all diverse perspectives are reasonably well-captured and included in the summarized results.

C Simulation Replicability and Reproducibility

Readers can run the simulations and validation scripts described in this paper by navigating to the project GitHub repository. Using the terminology from Brodeur et al. (2024), running the simulator with the script's preset parameter values will *replicate*, but not *reproduce*, the results we report in the paper. We provide the data and code in our reproduction GitHub repository so that readers can reproduce the paper's figures using our simulated results. Our reproduction repository shows we can reproduce the results reported in the paper from the raw data that resulted from our simulation runs.

As Spirling (2023) notes, it is not possible to reproduce the simulation output since we use a proprietary model (Gemini) as our primary LLM. Further, Google and Groq deprecate or change the LLMs from time-to-time and we have no control over that. For these reasons, users cannot replicate the exact simulation results we report in the paper. The core issue for reproducibility with LLM research is that the data generating process of the LLM is not only unknown, but is also subject to change at arbitrary times by the company that controls it. Reproduction in this context is analogous to analyzing administrative data provided by a government agency that does not disclose its process for assembling data and that changes data collection and curation procedures from time to time. It also is analogous to survey data for a geographic region where the population is mobile and the population composition is subject to change.

However, since we focus on the distribution patterns of cosine similarity scores rather than point estimates for both the main results and the bias evaluation, the replicated results should be similar to the results we report in the paper. And indeed, after we completed the experiments for this paper in late May 2026, Groq discontinued two of the models we relied on, Qwen3 32B and Llama 3.3 70B versatile. As a result, we conducted a complete replication of all results reported in this paper but replacing these two models with llama-3.1-8b-instant, and still using Gemini 3.0 flash preview and OpenAI GPT OSS 120B. In this replication, half of the models are different, and since two months elapsed, presumably Google made changes to Gemini as well. The results of this replication are shown in Tables A1-A4. Note that for the $N = 600$ design, the means and standard deviations of the cosine similarity results are virtually identical in the replication, and nearly so for the $N = 60$ design. The $N = 6$ design performs less well in the replication, showing that large sample sizes for the trials also improves the replicability of the results.

D Demonstrating the Generality of the Methods

The project GitHub repository contains all of the open source Python scripts needed to run the digital twin simulator for the structured group design, along with detailed instructions. To run the simulations, one simply needs to modify the main script to specify the topic, standpoint options and background information to a topic and question of interest, the covariates for the personas, and run the script locally. One can leave the scripts at their default values to replicate the results we report.

The methods we describe in this paper can apply to any synchronous research design. However, to assess a new design, a new Python script for the simulator would need to be written. The author team collaborated to write the existing Python scripts by hand

Table A1: Results for Nuclear Power, 4 Options, Random Sampling

	Empty	**Not Empty**		
	Rate	Mean	SD	N
N=6				
Expand	**0.10**	**0.69**	**0.11**	**990**
	0.08	0.70	0.10	1035
Maintain	**0.48**	**0.63**	**0.14**	**325**
	0.50	0.63	0.11	300
Phase Out	**0.62**	**0.74**	**0.08**	**171**
	0.58	0.67	0.12	210
Research	**0.02**	**0.74**	**0.09**	**1176**
	0.02	0.72	0.11	1176
N=60				
Expand	**0**	**0.84**	**0.04**	**435**
	0	0.83	0.04	435
Maintain	**0**	**0.78**	**0.05**	**435**
	0	0.78	0.05	435
Phase Out	**0**	**0.80**	**0.05**	**435**
	0	0.79	0.04	435
Research	**0**	**0.81**	**0.04**	**435**
	0	0.81	0.05	435
N=600				
Expand	**0**	**0.86**	**0.03**	**190**
	0	0.84	0.03	190
Maintain	**0**	**0.84**	**0.03**	**190**
	0	0.84	0.03	190
Phase Out	**0**	**0.84**	**0.04**	**190**
	0	0.83	0.04	190
Research	**0**	**0.85**	**0.03**	**190**
	0	0.86	0.03	190

Bold Font: Original, Standard Font: Replication

Table A2: Results for Nuclear Power, 3 Options, Random Sampling

	Empty	**Not Empty**		
	Rate	Mean	SD	N
N=6				
Expand	**0**	**0.76**	**0.07**	**1125**
	0.02	0.71	0.09	1176
Maintain	**0.24**	**0.69**	**0.12**	**715**
	0.16	0.66	0.11	861
Phase Out	**0.24**	**0.76**	**0.07**	**666**
	0.16	0.69	0.12	861
N=60				
Expand	**0**	**0.84**	**0.04**	**435**
	0	0.84	0.04	435
Maintain	**0**	**0.82**	**0.03**	**435**
	0	0.81	0.05	435
Phase Out	**0**	**0.82**	**0.04**	**435**
	0	0.83	0.04	435
N=600				
Expand	**0**	**0.85**	**0.03**	**190**
	0	0.85	0.03	190
Maintain	**0**	**0.85**	**0.04**	**190**
	0	0.85	0.04	190
Phase Out	**0**	**0.86**	**0.03**	**190**
	0	0.87	0.03	190

Bold Font: Original, Standard Font: Replication

Table A3: Results for Nuclear Power, 3 Options, Individual Design

	Empty	**Not Empty**		
	Rate	Mean	SD	N
N=6				
Expand	**NA**	**NA**	**NA**	**NA**
	0.02	0.71	0.11	1177
Maintain	**NA**	**NA**	**NA**	**NA**
	0.12	0.71	0.08	946
Phase Out	**NA**	**NA**	**NA**	**NA**
	0.16	0.74	0.08	861
N=60				
Expand	**NA**	**NA**	**NA**	**NA**
	0	0.86	0.03	435
Maintain	**NA**	**NA**	**NA**	**NA**
	0	0.82	0.04	435
Phase Out	**NA**	**NA**	**NA**	**NA**
	0	0.83	0.04	435
N=600				
Expand	**NA**	**NA**	**NA**	**NA**
	0	0.89	0.02	190
Maintain	**NA**	**NA**	**NA**	**NA**
	0	0.87	0.03	190
Phase Out	**NA**	**NA**	**NA**	**NA**
	0	0.87	0.03	190

Bold Font: Original, Standard Font: Replication

Note: This experiment was not run on the original models.

Table A4: Results for Gun Control, 3 Options, Stratified Sampling

	Empty	**Not Empty**		
	Rate	Mean	SD	N
N=6 (Random)				
More Strict	**0**	**0.80**	**0.06**	**1225**
	0.02	0.72	0.09	1177
About Right	**0.36**	**0.57**	**0.12**	**496**
	0.12	0.58	0.13	964
Less Strict	**0.24**	**0.66**	**0.12**	**703**
	0.32	0.62	0.11	561
N=6 (Stratified)				
More Strict	**0**	**0.77**	**0.06**	**1225**
	0	0.76	0.07	1225
About Right	**0.28**	**0.61**	**0.10**	**630**
	0.34	0.56	0.12	568
Less Strict	**0.02**	**0.71**	**0.09**	**1176**
	0.04	0.69	0.11	1128
N=60 (Random)				
More Strict	**0**	**0.84**	**0.04**	**435**
	0	0.86	0.03	435
About Right	**0**	**0.80**	**0.05**	**435**
	0	0.81	0.04	435
Less Strict	**0**	**0.81**	**0.05**	**435**
	0	0.83	0.04	435

Bold Font: Original, Standard Font: Replication

for each of the simulations and each of the validation tests we have described so far. However, to show the ease by which these methods can be applied to new designs, we used Claude Code to create two new simulators. First, we used Claude to create the no-summary version of the structured group design. And we used Claude to create what we call the *unstructured group design*. We requested Claude to build these scripts by first examining the existing structured design simulator script, and then modifying the script to accommodate these changes. Specifically, to create the unstructured version it modified the structured design script by removing the closed-ended response options, and instead only have the personas brainstorm ideas on the topic through a multi-round discussion. It created the no-summary version by sending persona randomly selected individual responses instead of summarized responses.

These new scripts show generality of our simulation methods by having Claude modify the script to implement a different synchronous design given all of the dependencies in code. The new script for the unstructured design simulator only required two requests from Claude, an initial request and then a simple correction, and it took only minutes to complete. The unstructured version worked first time off the shelf. The possibilities for alternative designs are endless. For example, an extension for the unstructured design could be to use a stance extraction tool to assign opposite-stance reasons across the rounds to create discussion across differences for the unstructured group design.

The unstructured design simulator is available at our project GitHub repository. More generally, this synchronous digital twin simulation project is fully open source and the code is accessible at our repository. We encourage anyone who is interested to join our open source project as a contributor to help build a library of alternative designs.

E AI Disclosure

Here we disclose the use of AI for this project.

What AI was used for. We used AI in five ways for this project. First, as we describe explicitly in the main text, we used LLMs in the simulation, analysis and validation pipelines. The simulator uses covariate data to construct LLM prompts, which in turn create an ensemble of LLM "personas" that are tasked with deliberating an issue. The simulator also uses an LLM prompt to summarize the content that the ensemble of personas generate. For the main analysis as well as our bias audit, we convert the LLM summarized text to embeddings in order to implement cosine similarity metrics. For validation, we make use of an LLM to classify persona-generated content by topic. Each aspect of the simulation, analysis and validation pipelines, and how we use LLMs for each, is explained in detail and explicitly in the manuscript.

Second, we made use of an LLM to streamline the Python script for the core structured group design simulator. To do this streamlining, we asked the LLM to read the human-written version of the simulator, and we asked it to create a new version of the code that works (exactly) identically but is more efficiently coded. All of the underlying logic of the simulator remains human-written, but the code in the updated version is a bit more modular (and the LLM had a preference for using data classes for storage instead of dictionaries that we had used). We disclose this use of the LLM in the acknowledgments section of the paper and in the code repository.

Third, as we describe explicitly in this appendix, we made use of an LLM to create new Python scripts for the non-summarized structured group design and the unstructured group design. Using the LLM to create the new unstructured design script was an experiment to demonstrate that the code repository is easily extensible to other synchronous designs. We only report that we conducted this experiment in code generation but we do not report results from this LLM-generated simulator.

Fourth, we made use of an LLM to write the code to create figures 1, 2 and A1-A4. This code is located in the project reproduction repository.

Fifth, we made use of an LLm to do a peer review of the initial draft of the paper.

We provide the text of this peer review, as well as our responses to the peer review and a listing of the revisions this peer review prompted, in the project replication repository.

Which tools were used. To generate simulated data among our LLM personas, we make use of four different LLMs: Gemini 3.0 flash preview, Qwen3 32B, Llama 3.3 70B versatile, and OpenAI GPT OSS 120B, with the latter three being hosted on Groq and are open weight LLMs. To generate the summaries we use Gemini 3.0 flash preview. We state these models in footnote 11. To create embeddings for the analysis, we use the Hugging Face sentence transformer all-MiniLM-L6-v2. We state this model in footnote 13. To do the coding streamline project and the new script generation, we use Claude Sonnet 4.6. We also state this model in the acknowledgment section. We used Claude Sonnet 5 for creating the figure code and for the peer review.

How human oversight was maintained. A core aspect of our paper is the development and implementation of validation strategies for LLM-based discourse simulation. We describe our validation methods in sections 5 and 6 and in the appendix of the manuscript. This includes our proposals for backward continuity, frontward continuity, and coverage, each of which required a research assistant to review the LLM output for accuracy, and then the full lab met to review the research assistant's work. Finally, for the code creation and streamlining tasks, the PI read the code line by line to ensure that code was exact and correct.

Table A5: Summaries for Nuclear Power Focus Group

	Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction
Stage 1: Initial Arguments	
Option 1	Nuclear power is viewed as a reliable, low-carbon energy source necessary for meeting climate goals and reducing reliance on fossil fuels. Energy independence and security are seen as benefits of nuclear power, reducing dependence on foreign sources and unreliable renewables. Technological advancements in safety and waste management are believed to mitigate concerns associated with nuclear power. Nuclear power is considered a viable solution for meeting increasing energy demands while providing economic benefits such as job creation. Nuclear power is seen as a practical solution for transitioning away from fossil fuels, especially when considering the limitations of intermittent renewable energy sources.
Option 2	A balanced approach is favored, as it allows for the continued use of a significant energy source without the perceived risks of rapid expansion or the negative impacts of premature phase-out. The status quo is seen as a pragmatic solution, given existing political, social, and environmental complexities, as well as the desire to avoid further polarization while utilizing current infrastructure. Concerns regarding the risks associated with nuclear waste and potential accidents are considered significant enough to warrant caution, emphasizing the need for further advancements in safety and waste management before any expansion is considered. Maintaining current nuclear operations allows for a reliable, low-carbon energy source to be preserved, while further research into safer and more sustainable energy solutions is conducted. The current approach is viewed as a responsible middle ground, balancing energy needs with environmental and safety concerns, and allowing for continued research and public discourse before committing to significant changes.
Option 3	The dangers of long-term nuclear waste storage and the potential for catastrophic accidents are considered unacceptable risks. Investment in renewable energy sources is believed to be a more environmentally responsible and sustainable alternative. The risks associated with nuclear energy are viewed as outweighing any low-carbon benefits. Concerns about the safety and potential environmental impact of nuclear energy are prioritized over its advantages.

Table A1: Summaries for Nuclear Power Focus Group (cont.)

	Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction
Stage 2: Counterarguments	
Option 1	Long-term safety and waste disposal concerns are seen as unresolved, with the potential for accidents and mismanagement outweighing the benefits, regardless of technological advancements. The high costs, long construction times, and complex supply chains associated with nuclear power are viewed as potential impediments to a rapid energy transition, diverting resources from more readily available and resilient renewable solutions. The necessity of overhauling the existing energy infrastructure to prioritize nuclear power is questioned, particularly when current needs are being met and other renewable options are available. Prioritizing nuclear power for energy independence is believed to create new vulnerabilities and distract from investments in domestic renewable energy sources. The potential long-term risks associated with rapid nuclear expansion, including waste disposal and safety concerns, are considered to outweigh the economic benefits of job creation and meeting energy demands.
Option 2	A reliance on fossil fuels is seen to be prolonged by not expanding nuclear power, potentially missing critical climate targets due to the limitations of renewable energy sources alone. The risks associated with nuclear power, such as waste disposal and potential accidents, are considered to outweigh the short-term benefits, with a preference for prioritizing a rapid transition to renewable energy sources. A delay in scaling up nuclear power is viewed as a risk to energy security and climate action, as nuclear is currently seen as a readily available solution, and scaling up is preferred while pursuing other innovations. Maintaining the current status of nuclear operations is considered insufficient for achieving necessary emission reductions, and a more proactive approach involving new construction is seen as necessary to displace fossil fuels and meet energy demands. Continuing current nuclear operations is seen as delaying the transition to truly sustainable and secure energy solutions, with concerns that it perpetuates reliance on outdated systems and diverts resources from renewable alternatives.
Option 3	The reliable baseload power provided by nuclear energy is considered essential for grid stability and meeting energy demands, especially when renewables are intermittent or not yet scalable. Modern advancements in nuclear technology and stringent safety protocols are believed to have significantly mitigated risks associated with nuclear energy, making it

Table A1: Summaries for Nuclear Power Focus Group (cont.)

	Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction
(cont.)	a safe and manageable energy source. The risks associated with nuclear energy, including waste management and potential accidents, are viewed as manageable with proper technology and regulation, and are outweighed by the existential threat of climate change and the harms of fossil fuels. Nuclear energy is seen as a critical component in transitioning to a low-carbon economy and achieving energy independence, offering a proven, scalable, and low-carbon alternative to fossil fuels without the ecological trade-offs of renewables. Prematurely phasing out existing nuclear plants is viewed as counterproductive, potentially leading to increased reliance on fossil fuels and hindering progress towards climate goals, as well as wasting existing infrastructure.
Stage 3: Responses to Counterarguments	
Option 1	Nuclear power is believed to be a necessary component of a diversified energy portfolio, offering a reliable baseload power source that complements intermittent renewable energy sources and reduces reliance on fossil fuels. Concerns about the safety and waste disposal associated with nuclear power are acknowledged, but it is argued that advancements in reactor designs, waste management technologies, and regulatory oversight have significantly mitigated these risks, making them manageable. The expansion of nuclear power is considered an investment in long-term energy independence, providing a stable and secure energy supply that reduces vulnerability to volatile global markets and geopolitical instability. The urgency of addressing climate change is emphasized, with nuclear power viewed as a crucial tool for decarbonizing the energy sector quickly and effectively, especially given the limitations of renewables in meeting baseload power demands. Modernizing infrastructure and investing in nuclear power is seen as a means of future-proofing the energy grid, preparing for increased energy demands and ensuring a sustainable energy supply for future generations.

Table A1: Summaries for Nuclear Power Focus Group (cont.)

	Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction
Stage 3: Responses to Counterarguments (cont.)	
Option 2	A rapid expansion of nuclear power is viewed as unnecessary, as renewable energy sources are believed to have the potential for further development and could meet energy needs. Maintaining current nuclear operations is seen as a pragmatic approach to ensure a stable energy supply while improvements to waste management and safety technologies are made and renewables are scaled up. Decommissioning existing nuclear plants before renewables are fully capable of meeting demand is viewed as risky, potentially creating energy instability and increasing reliance on fossil fuels. Building new nuclear plants is seen as a potentially risky and costly endeavor, with concerns raised about waste disposal, long-term risks, and the prioritization of renewable energy sources. A cautious approach to nuclear energy is favored, focusing on optimizing existing plants, improving safety, and exploring the potential of renewable energy sources before committing to large-scale new construction.
Option 3	The long-term risks associated with nuclear power, such as waste disposal and potential accidents, are believed to outweigh the benefits, especially when renewable energy sources are available. Reliance on nuclear power is viewed as unnecessary due to the availability of cleaner, safer, and more sustainable alternatives like solar and wind power. Concerns are raised that the term manageable does not equate to safe, particularly in the context of nuclear power's potential for catastrophic accidents and long-term environmental impact. The possibility of a short-term energy gap resulting from phasing out nuclear plants is acknowledged, but it is suggested that this gap can be bridged with investments in renewable energy and storage solutions. Continued investment in nuclear power is seen as diverting resources from the development and deployment of renewable energy technologies.

Table A2: Summaries for Expanding House Focus Group – Human

	Discussion Options: 1 = Increase the size of the U.S. House of Representatives by adding more seats 2 = Keep the size of the U.S. House of Representatives the same as it is now
Human Arguments	
Option 1	Smaller districts are seen as a way to foster closer relationships between representatives and constituents, ensuring **local** [**B1**] concerns and **diverse** [**B2**] voices are more accurately heard. It is believed that expansion will counteract **gerrymandering** [**B3**] and population disparities, leading to more equitable representation across different states and demographic groups. A larger legislative body is viewed as a means to distribute workload more effectively and provide the necessary resources to improve the quality and **efficiency** [**B4**] of governance. Increasing the number of representatives is expected to enhance accountability to the public while diluting the influence of **special interests** [**B5**] and **concentrated** [**B6**] political power. The expansion is considered a logical and necessary modernization of an outdated system to better reflect the current size and **complexity** [**B7**] of the American population.
Option 2	Concerns are raised regarding the increased financial burden on **taxpayers** [**A1**], including the costs of additional salaries, staff, and general government overhead. It is argued that a larger legislative body would become **unwieldy** [**A2**] and ineffective, leading to increased gridlock, confusion, and difficulty in reaching a consensus. The current number of representatives is viewed as sufficient, with many suggesting that the focus should be on improving the performance and **accountability** [**A3**] of existing members rather than expanding the government. Fears are expressed that increasing the size of the House would be used for **partisan** [**A4**] gain, exacerbate issues like **gerrymandering** [**A5**], and potentially increase political corruption. It is suggested that modern communication **technology** [**A6**] and **existing** [**A7**] Constitutional structures make the current level of representation adequate, rendering an expansion unnecessary.

Table A3: Summaries for Expanding House Focus Group – Simulation, Stage 1

	Discussion Options: 1 = Increase the size of the U.S. House of Representatives by adding more seats 2 = Keep the size of the U.S. House of Representatives the same as it is now
LLM Arguments	
Option 1	Accountability and responsiveness are seen as being enhanced through smaller districts that foster **closer** [**B1**], more direct connections between representatives and the people they serve. More equitable representation for a growing and **diverse** [**B2**] population is believed to be secured by preventing individual and marginalized voices from being diluted in large districts. Potential increases in government spending and administrative complexity are viewed as being outweighed by the democratic necessity of a more participatory and granular legislative body. The influence of **entrenched** [**B6**] political elites and rigid hierarchies is thought to be diluted by decentralizing power and making representatives more directly accountable to local communities. More effective and nuanced policymaking is expected to be achieved through a better understanding of unique community needs and increased local engagement.
Option 2	Increased financial burdens on **taxpayers** [**A1**] and federal budget expansions are cited as primary reasons to avoid adding more seats. The maintenance of the current size is seen as necessary to prevent the legislative process from becoming more complicated, inefficient, **unwieldy** [**A2**], and bureaucratic. The preservation of established institutional structures, hierarchical order, and trust in existing leadership is prioritized to ensure **stable** [**A7**] governance. Potential improvements in constituent connection or personal impact are viewed as **unproven** [**LLM only**] or insufficient to justify the associated costs and risks of instability.

Table A4: Summaries for Expanding House Focus Group – Simulation, Stage 2

	Discussion Options: 1 = Increase the size of the U.S. House of Representatives by adding more seats 2 = Keep the size of the U.S. House of Representatives the same as it is now
LLM Arguments	
Option 1	It is argued that expansion is essential to ensure that a growing, diverse population and marginalized communities are adequately represented and heard within the legislative process. The financial impact is viewed as a negligible investment that is outweighed by the long-term savings and efficacy generated through more locally attuned, equitable policy-making. Potential administrative complexities are seen as manageable through modern organizational structures and specialized committees that would decentralize power and improve nuanced policy debate. Institutional legitimacy and public trust are believed to be strengthened when outdated, rigid hierarchies are evolved to be more accessible and responsive to the **modern** [**B7**] electorate. Smaller districts are favored as a means to foster greater legislative **accountability** [**B5**], increase voter engagement, and ensure that representatives are not spread too thin to address constituent needs.
Option 2	An increase in House size is viewed as an unnecessary expansion of the federal bureaucracy that would lead to excessive government spending, administrative waste, and an increased financial burden on taxpayers. Expansion is seen as a catalyst for legislative gridlock and organizational chaos, which would fragment national representation and make it significantly harder to reach consensus or take decisive action. It is argued that adding seats would dilute the significance of individual representatives and decrease **accountability** [**A3**], potentially leading to a less experienced legislative body that is more reliant on centralized party leadership. The current legislative framework is considered sufficient for addressing community needs, with existing tools and established traditions providing a more effective balance of representation and stability than a larger, more complex system.

Table A5: Summaries for Expanding House Focus Group – Simulation, Stage 3

	Discussion Options: 1 = Increase the size of the U.S. House of Representatives by adding more seats 2 = Keep the size of the U.S. House of Representatives the same as it is now
LLM Arguments	
Option 1	It is argued that an increase in seats is necessary to ensure that diverse, marginalized, and local voices are accurately heard, as smaller districts allow for more personalized and responsive representation in a growing population. The expansion is viewed as a way to increase direct accountability to constituents and dismantle the influence of centralized party elites by making representatives more accessible and answerable to manageable local communities. The administrative costs of adding seats are considered a modest and necessary investment in democratic legitimacy, with the potential for long-term savings through more **effective** [**B4**], community-focused decision-making and reduced bureaucratic waste. It is suggested that a larger House could improve legislative quality and break current gridlock by fostering specialized expertise, creative problem-solving, and the formation of more **nuanced** [**LLM only**], issue-specific coalitions. The preservation of an outdated legislative structure is seen as a risk to stability, necessitating an evolution of the House's size to reflect modern population dynamics and ensure the system remains functional and equitable.
Option 2	Expansion is viewed as a risk that would create a bloated, fragmented bureaucracy, leading to increased legislative gridlock and slower decision-making processes. Significant fiscal costs and an increased tax burden are seen as unjustifiable, as the additional spending is not expected to yield a more efficient or effective government. The long-standing, traditional structure of the House is regarded as a proven and stable system that should be preserved to avoid the unnecessary complications and risks of altering a functioning institution. Effective representation is believed to be better achieved through improving the quality of leadership and constituent engagement within the existing framework rather than simply adding more seats. The potential for increased complexity and diluted legislative effectiveness is seen as a major drawback, as a smaller, more manageable body is thought to maintain better institutional integrity and **accountability** [**A3**].